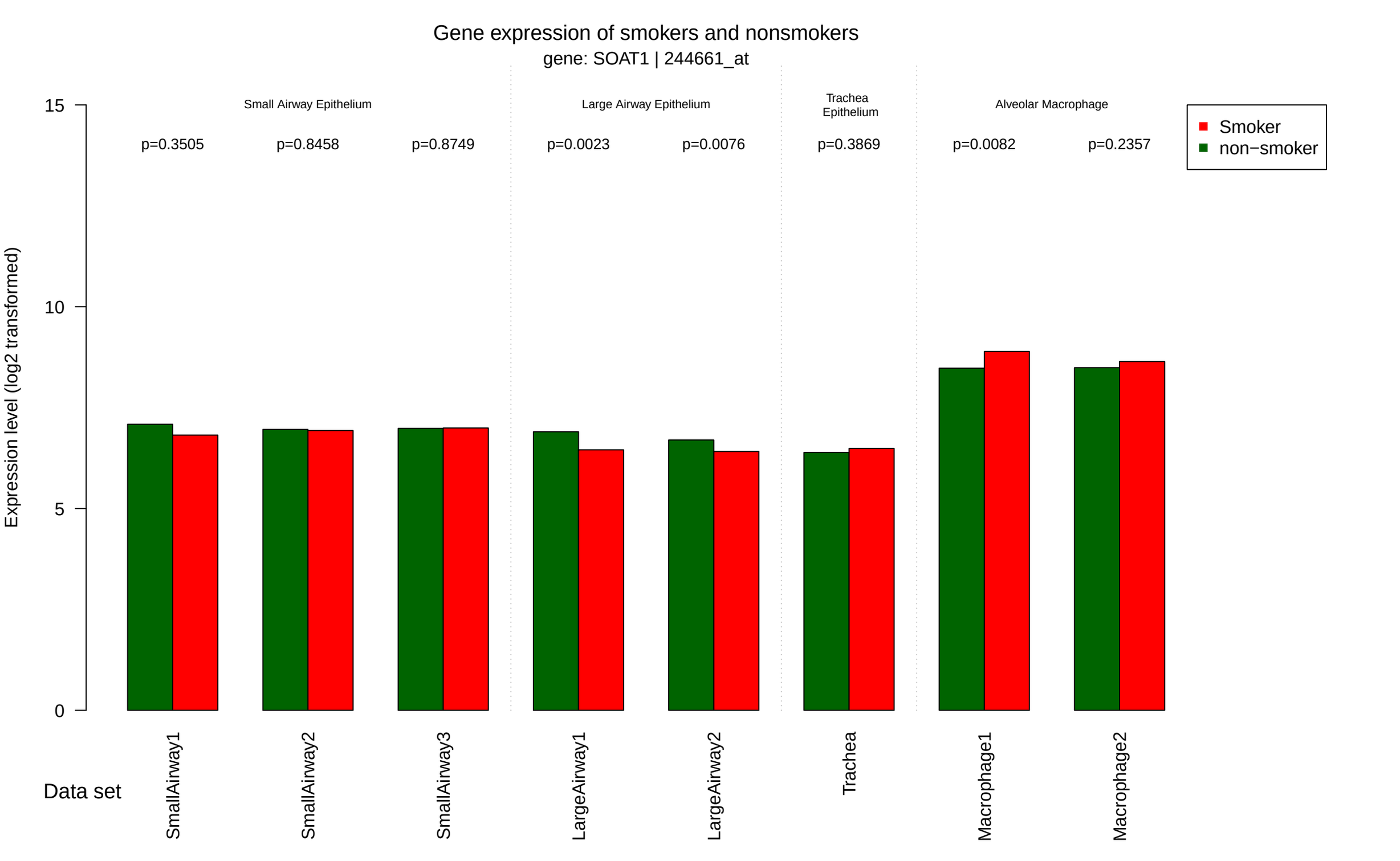
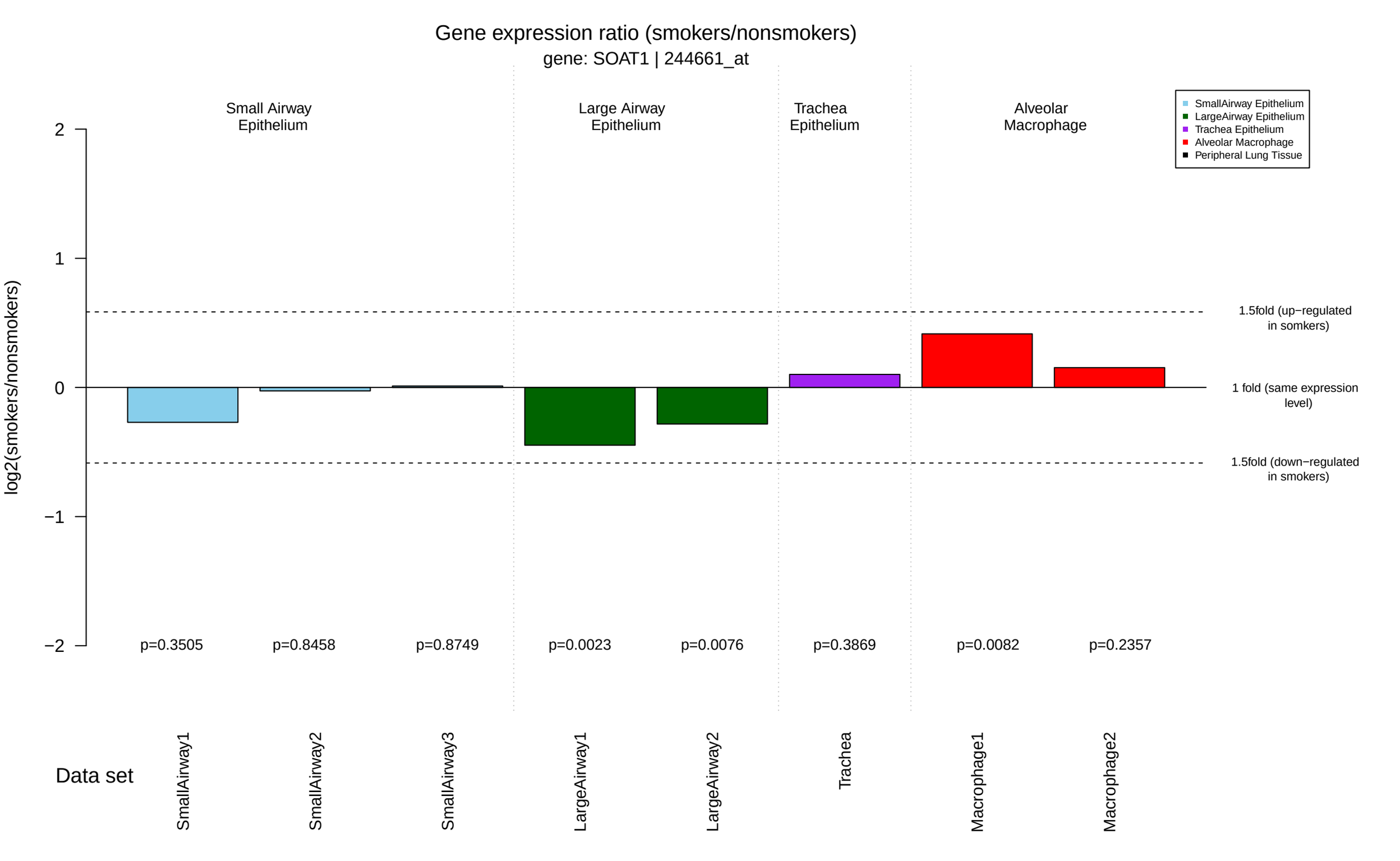
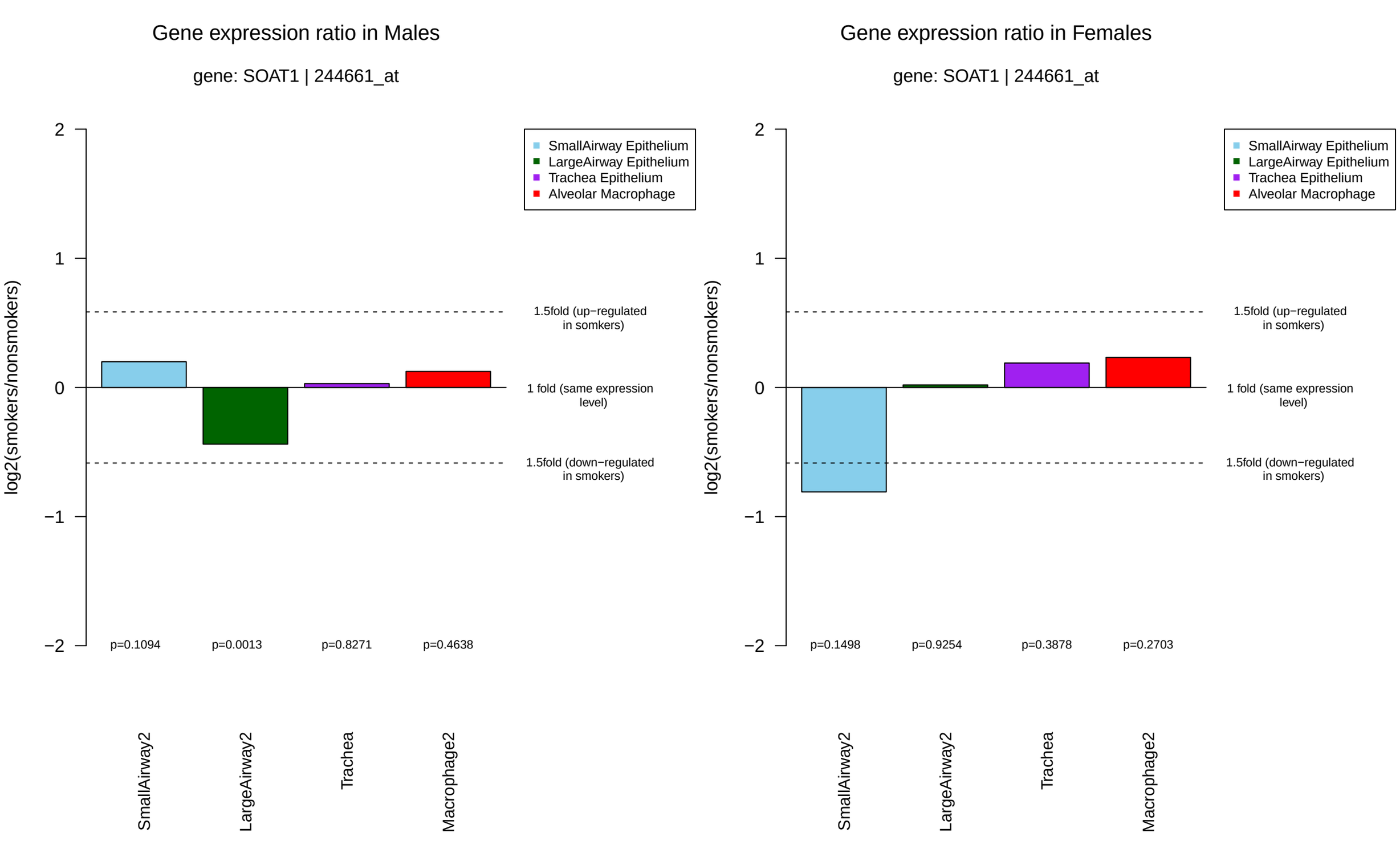
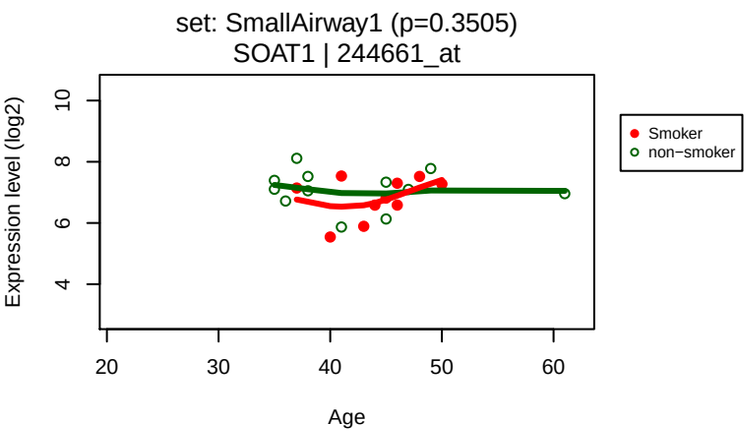
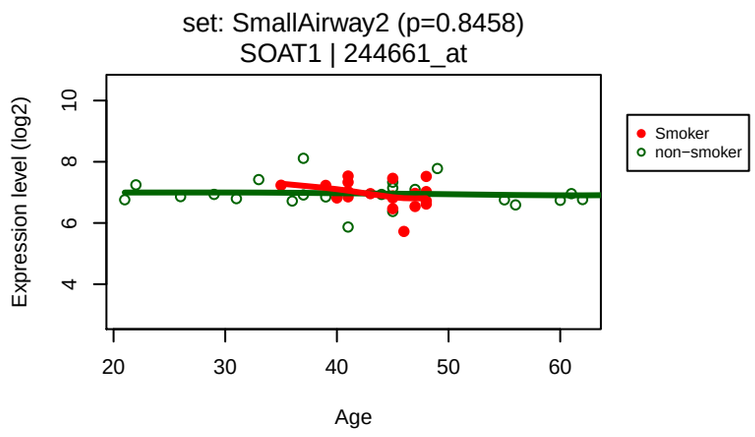
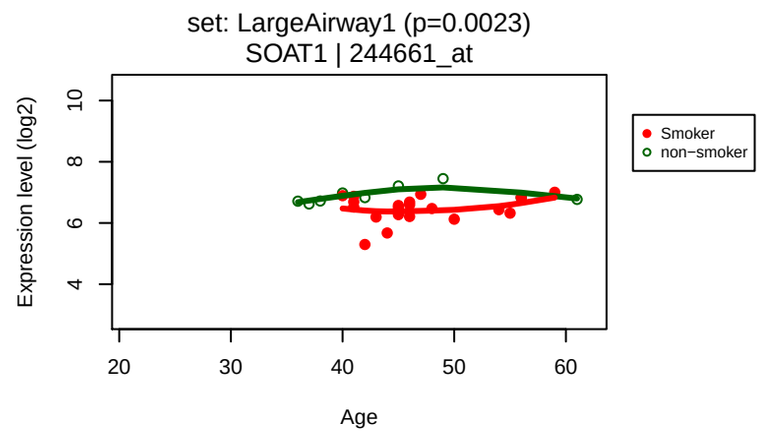

Gene expression of smokers and nonsmokers
gene: SOAT1 | 244661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3505
p=0.8458
p=0.8749
p=0.0023
p=0.0076
p=0.3869
p=0.0082
p=0.2357
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

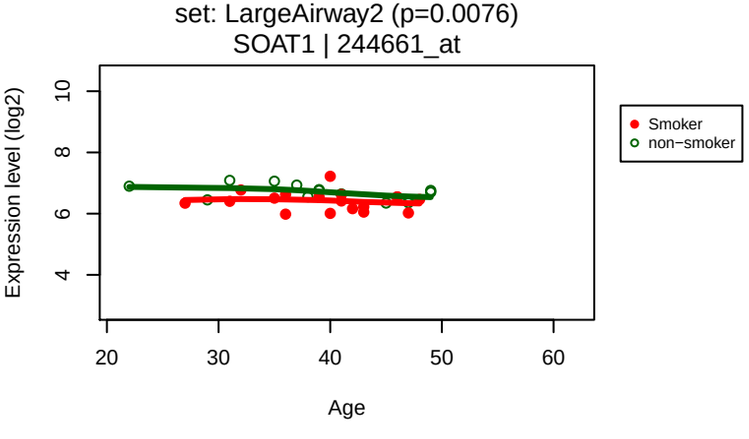

Gene expression ratio (smokers/nonsmokers)
gene: SOAT1 | 244661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3505
p=0.8458
p=0.8749
p=0.0023
p=0.0076
p=0.3869
p=0.0082
p=0.2357
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

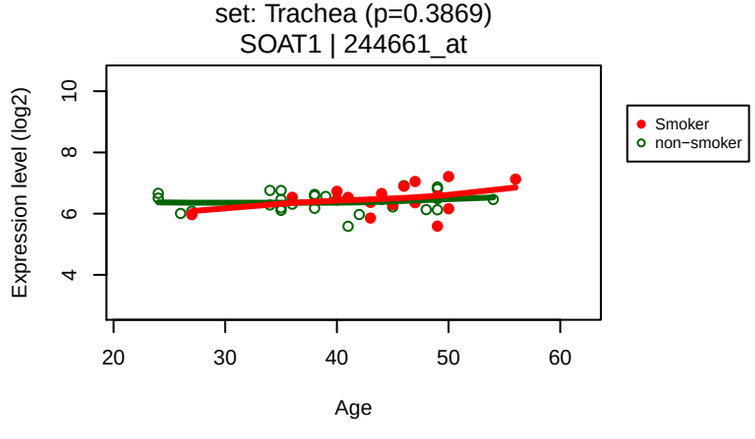

Gene expression ratio in Males
gene: SOAT1 | 244661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1094
p=0.0013
p=0.8271
p=0.4638
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOAT1 | 244661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1498
p=0.9254
p=0.3878
p=0.2703
SmallAirway2
LargeAirway2
Trachea
Macrophage2

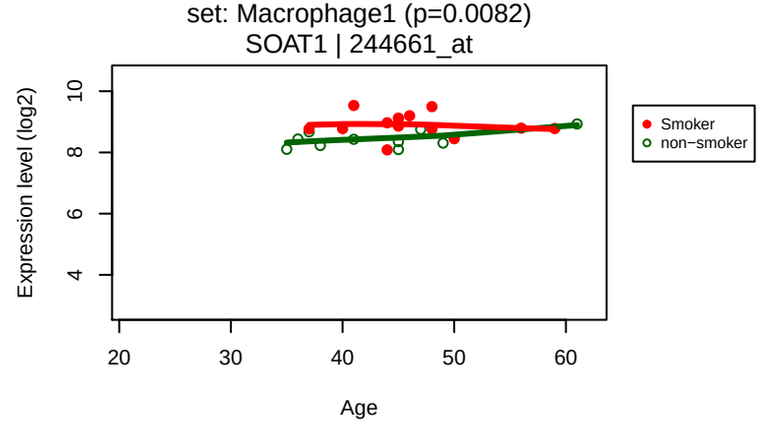

set: SmallAirway1 (p=0.3505)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker

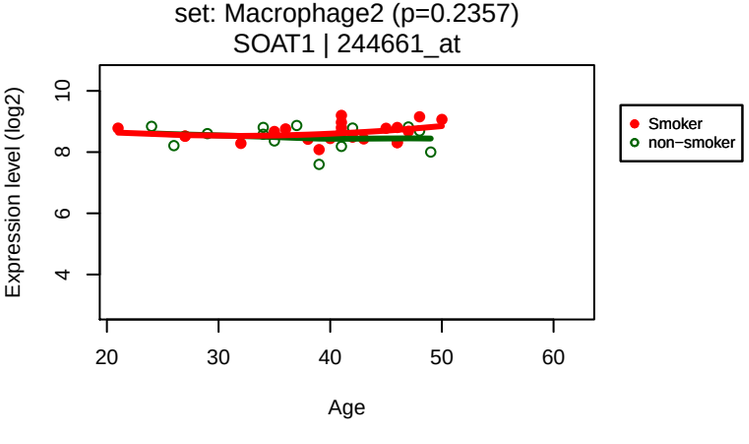

set: SmallAirway2 (p=0.8458)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker

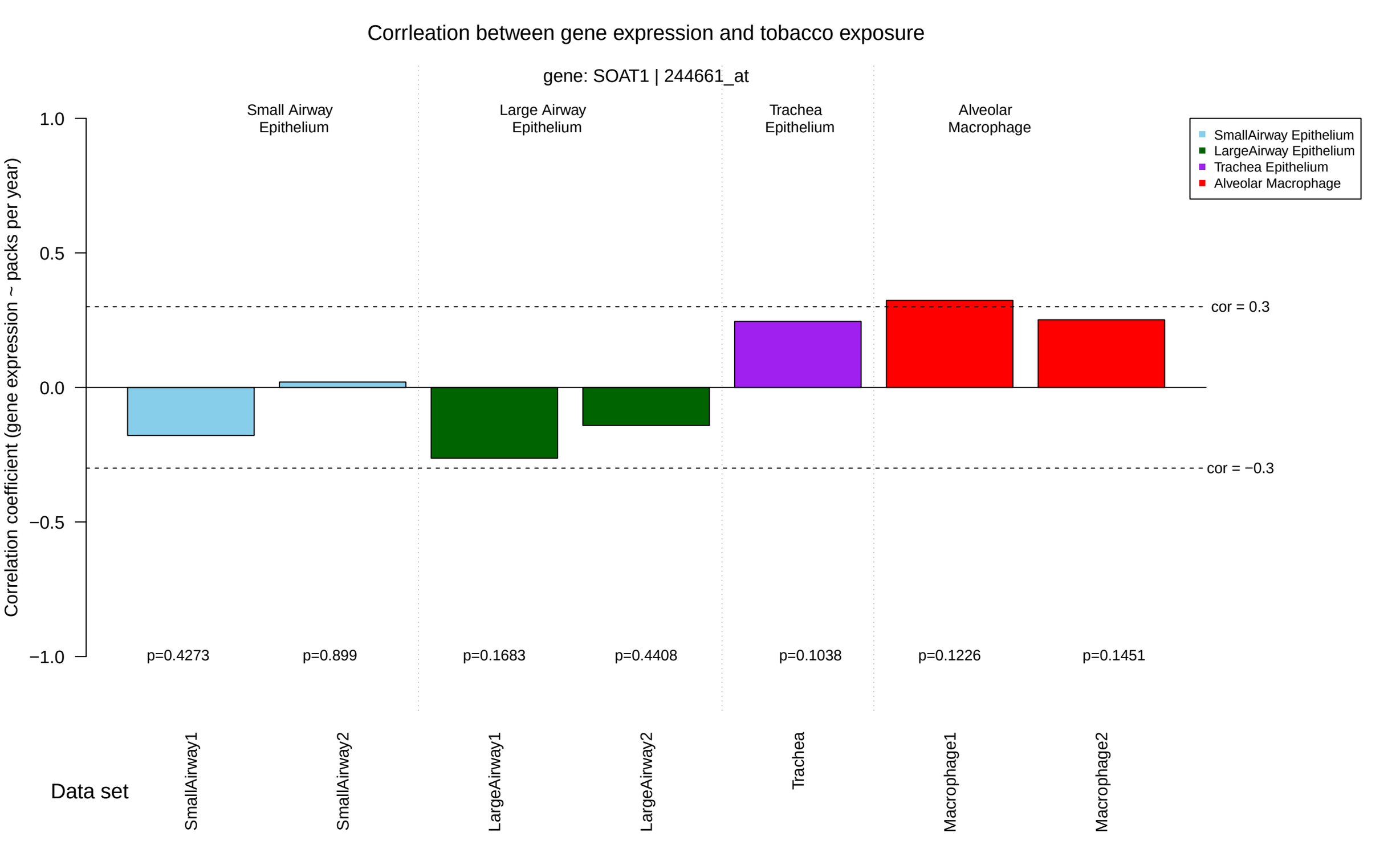

set: LargeAirway1 (p=0.0023)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0076)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3869)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0082)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2357)
SOAT1 | 244661_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOAT1 | 244661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4273
p=0.899
p=0.1683
p=0.4408
p=0.1038
p=0.1226
p=0.1451
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2